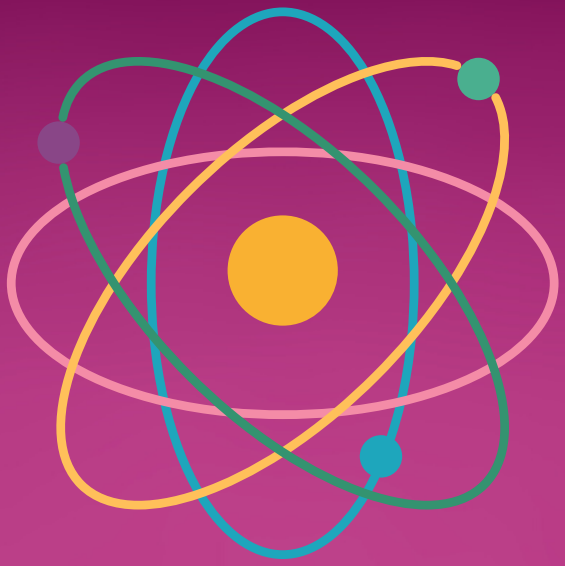
ALLEN CAREER INSTITUTE
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PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Thermodynamics-02

English Medium

EXERCISE # S-I

❑ Carnot Cycle

1. The efficiency of a carnot cycle is 1/6. On decreasing the temperature of the sink by 65°C, the efficiency increases to 1/3. Calculate the temperature of source and sink. **TD0001**

2. A carnot cycle has an efficiency of 40%. Its low temperature reservoir is at 9°C. What is the temperature of source ? **TD0002**

3. A heat engine absorbs 760 kJ heat from a source at 380K. It rejects (i) 650 kJ, (ii) 560 kJ, (iii) 504 kJ of heat to sink at 280K. State which of these represent a reversible, an irreversible or an impossible cycle. **TD0115**

4. A carnot engine operating between 227°C and 27°C absorbs 2 Kcal of heat from the 227°C reservoir reversibly per cycle. The amount of work done by engine in one cycle is (in kcal):- **TD0116**

❑ Entropy

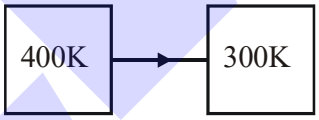
5. A diatomic ideal gas is expanded isothermally to 32 times of it's initial volume then it is cooled to restore to initial entropy at constant volume. Calculate ratio of intial temperature to final temperature **TD0003**

6. One mole of an ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate ΔS_{system}, ΔS_{surr} and ΔS_{total}

(i) when the process is carried out reversibly

(ii) when the process is carried out irreversibly (one step) **TD0004**

7. 100 kJ heat is transferred from a large heat reservoir at 400 K to another large heat reservoir at 300 K. Suppose there is no change in temperature due to exchange of heat and combinedly source and sink form isolated sytem.



Find (a)ΔS_{source}, (b) ΔS_{sink} and (c) ΔS_{total}. Also comment on spontaneity of process. **TD0005**

8. Find the change in entropy (in cal/K) of 1 mole of O_2 gas (C_V = 5/2 R), when it is

(a) heated from 300 K to 400 K isobarically

(b) heated from 300 K to 400 K isochorically (Given : ln 3 = 1.1, ln2 = 0.7)

TD0117

9. The entropy change (in J/K) involved in the isothermal reversible expansion of 2 moles of an ideal gas from a volume of 10 dm^3 to a volume of 100 dm^3 at 27°C is :- **TD0118**

10. One mole of NaCl(s) on melting as it's melting point absorbed 32.76 kJ of heat and its entropy is increased by 30 JK^{-1}. What is the melting point of sodium chloride (in °C) ? **TD0119**

11. A system of 100 kg mass undergoes a process in which its specific entropy increases from 0.3 $kJkg^{-1}K^{-1}$ to 0.4 $kJkg^{-1}K^{-1}$. At the same time, the entropy of the surrounding decreases from 80 kJK^{-1} to 75 kJK^{-1}. Find the $(\Delta S)_{universe}$ in kJK^{-1}. **TD0006**

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12. Calculate $(\Delta S)_{universe}$ (in cal/K) when 1kg water at 300K is brought into contact with a heat reservoir at 600K. Specific heat capacity of water is 1cal/gm-°C. Assume no change in physical state of water.

(ln2 = 0.7) **TD0007**

13. 1 mole of ideal monoatomic gas is heated by supplying 5 kJ heat from a reservoir maintained at 400 K from 300 K to 400 K. In the process volume of gas increased from 1 L to 10 L. Find ΔS_{total} (in J/K -mol) in the process

Use : $\ln\left(\frac{4}{3}\right) = 0.3$, ln 10 = 2.3, R = 8.3 J/K-mol and ln x = 2.3 log x **TD0008**

14. Calculate ΔS_r° at 298K of (in $JK^{-1}mol^{-1}$)

(i) $Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)$, **TD0009**

(ii) $\frac{1}{2}N_2(g) + 2H_2(g) + \frac{1}{2}Cl_2(g) \longrightarrow NH_4Cl(s)$ **TD0010**

(iii) $C(graphite) \longrightarrow C(diamond)$. **TD0011**

The values of S° of Na, Cl_2, NaCl, NH_4Cl, N_2, H_2 diamond & graphite are 51, 223, 72, 95, 192, 131, 2.43 & 5.69 JK^{-1} mol^{-1} respectively.

15. Idenfity the susbtance in each of the following pairs of samples that has the higher entropy?

(A) $Br_2(l)$ or $Br_2(g)$ (B) $C_2H_6(g)$ or $C_3H_8(g)$

(C) MgO(s) or NaCl(s) (D) KOH(s) or KOH(aq) **TD0012**

❑ **Gibb's Function**

16. Calculate the ΔG° (in kJ) change at 300 K for the reaction;

$Br_2(\ell) + Cl_2(g) \longrightarrow 2BrCl(g)$. For the reaction ΔH° = 29.3 kJ & the standard entropies of $Br_2(\ell)$, $Cl_2(g)$ & BrCl(g) at the 300 K are 150, 220, 240 J $mol^{-1}K^{-1}$ respectively. **TD0013**

17. The standard entropies of $H_2(g)$ and H(g) are 130 and 115 J $mol^{-1}K^{-1}$ respectively at 300K. Using the data given below calculate the bond energy of H_2(in kJ/mol) :-

$H_2(g) \longrightarrow 2H(g)$; ΔG° = 406 kJ/mol **TD0014**

18. 5 mole $H_2O(\ell)$ at 373K and 1 atm is converted into $H_2O(g)$ at 373K and 5 atm. Calculate ΔG (in Kcal) for this process. [Given : R = 2Cal/K-mol, ln 5 = 1.6] **TD0015**

19. Calculate ΔG (in bar-L) when a definite mass of a monoatomic ideal gas at 1 bar & 27°C is expanded adiabatically against vacuum from 10 L to 20 L (ln2 = 0.7) **TD0016**

20. Find $(\Delta S)_{universe}$ (in Joule/mole-K) at 1 bar for a chemical reaction at 300 K if $\Delta H^\circ_{300\,K}$ = 75 kJ / mol $\Delta S^\circ_{300\,K}$ = 300 J/K **TD0017**

21. A liquid freezes into a solid (ΔH = –1000 J/mole) at 200 K and 1 atm, (it's normal melting point).

(i) What is the value of ΔG (J/mole) at 200 K ?

(ii) What is the ΔS ($JK^{-1}mole^{-1}$) value at 200 K ?

(iii) Will the freezing be spontaneous at 150 K and 1 atm ? Calculate $\Delta S_{total.}$($JK^{-1}mole^{-1}$)

(iv) What is the value of ΔG ($J\,mole^{-1}$) at 250 K and 1 atm ?

(Assume that ΔS and ΔH do not depend on temperature) **TD0018**

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22. Animals operate under conditions of constant pressure and most of the processes that maintain life are electrical (in a broad sense). How much energy is available (in kJ) for sustaining this type of muscular and nervous activity from the combustion of 1 mol of glucose molecules under standard conditions at 37°C (blood temperature) ? The entropy change is + 210 JK^{-1} for the reaction as stated.

$\Delta H_{combustion\ [glucose]}$ = –2808 kJ / mol.

TD0120

23. Standard entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 $JK^{-1}mol^{-1}$, respectively. For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30$ kJ. For the given reaction to be in equilibrium absolute temperature will be ?

TD0121

❑ Thermodynamic and equilibrium constant

24. α-D Glucose undergoes mutarotation to β-D-Glucose in aqueous solution. If at 300K there is 60% conversion. Calculate ΔG° of the reaction. (ln2 = 0.7, ln 3 = 1.1)

α-D-Glucose $\rightleftharpoons$ β-D-Glucose **TD0019**

25. The equilibrium constant of the reaction $2C_3H_6(g) \rightleftharpoons C_2H_4(g) + C_4H_8(g)$ is found to fit the expression

$$\ln K = -1.04 - \frac{1088K}{T}$$

Calculate the standard reaction enthalpy in (kcal mol^{-1}) and entropy (in cal K^{-1} mol^{-1}) at 400 K.

TD0122

26. For the reaction,

$SO_2(g)$ + 1/2 $O_2(g)$ $\rightleftharpoons$ $SO_3(g)$; $\Delta H°_{300}$ = –95 kJ/mole,

$\Delta S°_{300}$ = –95.0 J/K mole. Find the value of (ln K_p) for this reaction at 300 K.

Give your answer as nearest integer. **TD0123**

27. For the reaction at 298 K

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$; $\Delta H° = -29.8$kcal; $\Delta S° = -0.1$ kcal/K

Calculate ΔG° (kcal/mole) and K_p for the reaction ? **TD0124**

28. For a dimerization reaction,

2 $A(g) \rightarrow A_2(g)$ at 300 K, ΔU^0 = – 20kJ mol^{-1}, ΔS^0 = –30 JK^{-1} mol^{-1}

Calculate the value of ΔG^0 (in kJ/mol). (R = 8.3 J/mole-K) **TD0125**

29. For the reaction ;

$X(g) \rightarrow 3Y(g)$; ΔG = – 2.7 kcal , ΔS = 20 cal K^{-1} at 300 K

Hence ΔU in kcal is_____ .

TD0126

30. The value of equilibrium constant of a reaction is given below :

Temperature	**Equilibrium constant**
T_1 = 25ºC	K_1 = 10

Find the value of ΔGº for the reaction at T_1 (in kJ mol^{-1}).

Give your answer as nearest integer.

TD0127

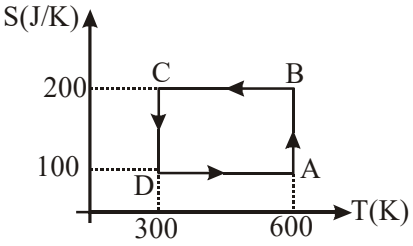
EXERCISE # S-II

1. From the given T-S diagram of a reversible carnot engine, find

(i) work delivered by engine in one cycle (in kJ) **TD0022**

(ii) heat taken from the source in each cycle.(in kJ) **TD0023**

(iii) ΔS_{sink} in each cycle. (in J/K) **TD0023**



2. At 300 K, $\Delta H^\circ_{combustion}$ (sucrose) $= -5000$ KJ/mol & $\Delta G^\circ_{combustion}$(sucrose) $= -6000$ KJ/mol. Estimate additional non-PV work (in kJ) that is obtained by raising temperature to 309 K.

Assume $\Delta_r C_P = 0$ for this temperature change. **TD0024**

3. The K_P for gaseous reaction $A + B \rightleftharpoons C + D$ is 2.0 at 27°C and 8.0 at 127°C. Determine the standard free energy change of this reaction at each temperature (in J/mole) and ΔH° (in kJ/mole) for the reaction over this range of temperature (ln2 = 0.7) ? **TD0025**

4. Pressure of 10 moles of an ideal gas is changed from 2 atm to 1 atm against constant external pressure without change in temperature. If surrounding temperature (300 K) and pressure (1 atm) always remains constant then calculate total entropy change ($\Delta S_{system} + \Delta S_{surrounding}$) (in J/K) for given process.

[Given : R = 8.3 J/mol-K, ln 2= 0.7] **TD0026**

5. What is $\Delta_r G$ (in kJ/mole) for synthesis of ammonia at 300 K at following sets of partial pressure

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; $\Delta_r G^\circ = -300$ kJ/mole.

[Take R $= \frac{25}{3}$ J/K mole, ln2 = 0.7, ln3 = 1.1, ln5 = 1.6]

Gas	N_2	H_2	NH_3
Pressure (atm)	1	3	0.02

TD0128

6. $2A(s) \rightleftharpoons B(g) + 2C(g) + 3D(g)$

Total pressure developed in a closed container by decomposition of A at equilibrium is 12 bar at 727ºC. Calculate $|\Delta G^\circ|$ (in kcal) of the reaction at 727ºC (R = 2cal/mole-K, ln 2 = 0.7, ln3 = 1.1)

TD0028

7. Find ΔG(J/mol) for the reaction at 300kPa & 27ºC when all gases are in stoichiometric ratio of moles.

$N_2O_4(g) \rightarrow 2NO_2(g)$ [ln2 = 0.7]

Given : $\Delta G_f^\circ(N_2O_4) = 100 kJ/mol$

$\Delta G_f^\circ(NO_2) = 50 kJ/mol$

R = 8 J/mol-K **TD0029**

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8. Oxygen is heated from 300 to 600 K at a constant pressure of 1 bar. What is the increase in its molar entropy (in J/K) ? The molar heat capacity (in JK^{-1} mol^{-1}) for the O_2 is (Given : $\ell n2 = 0.7$)

$C_{P,m} = 10 + 10^{-2}T$ **TD0030**

9. Calculate the molar entropy ((in JK^{-1} $mole^{-1}$)) of a substance at 600 K and 1 bar using the following data.

(i) Heat capacity of solid from 0K to normal melting point 200 K

$C_{P,m}(s) = 0.035\ T\ JK^{-1}mol^{-1}$.

(ii) Enthalpy of fusion = 7.5 kJ mol^{-1},

(iii) Enthalpy of vaporisation = 30 kJ mol^{-1},

(iv) Heat capacity of liquid from 200K to normal boiling point 300K

$C_{P,m}(\ell) = 60 + 0.016T\ JK^{-1}mol^{-1}$.

(v) Heat capacity of gas from 300K to 600K at 1 atm

$C_{P,m}(g) = 50.0\ JK^{-1}mol^{-1}$.

TD0129

10. For a reaction : $2A(s) + B(g) \longrightarrow 3C(l)$, standard entropy change of reaction is –2 kJ/mol-K and standard enthalpy of combustion of A(s), B(g) and C(l) are –100, –60, –285 kJ/mol respectively, then find the minimum work to be done at 27°C and 1 bar pressure to carry out the reaction (in kJ). **TD0130**

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EXERCISE # O-I

1. The maximum efficiency of a heat engine operating between 100°C and 25°C is :-

(A) 20.11% (B) 22.2% (C) 25.17% (D) None

TD0131

2. An ideal gas heat engine operates by carnot's cycle between 227ºC and 127ºC. It absorbs 6×10^4 cal of heat at high temperature. Amount of heat converted to work is :

(A) 2.4×10^4 cal (B) 4.8×10^4 cal (C) 1.2×10^4 cal (D) 6.0×10^4 cal

TD0132

3. A reversible heat engine A (based on carnot cycle) absorbs heat from a reservoir at 1000 K and rejects heat to a reservoir at T_2. A second reversible engine B (based on carnot cycle) absorbs the same amount of heat as rejected by the engine A, from the reservoir at T_2 and rejects energy to a reservoir at 360K.

If the efficiencies of engines A and B are the same then the temperature T_2 is :-

(A) 680 K (B) 640 K (C) 600 K (D) 670 K **TD0033**

4. Which of the following is incorrect for a closed system in which an irreversible process is occuring?

(A) $Q_{sys} + Q_{surr} = 0$ (B) $W_{sys} + W_{surr} = 0$ (C) $\Delta U_{sys} + \Delta U_{surr} = 0$ (D) $\Delta S_{sys} + \Delta S_{surr} = 0$

TD0034

5. For conversion C(graphite) $\rightarrow$ C(Diamond), the ΔS is :-

(A) Zero (B) Positive

(C) Negative (D) Can not be predicted **TD0035**

6. Considering entropy (S) as a thermodynamic parameter, the criteria for the spontaneity of any process is :-

(A) $\Delta S_{system} - \Delta S_{surroundings} > 0$ (B) $\Delta S_{system} > 0$ only

(C) $\Delta S_{surroundings} > 0$ only (D) $\Delta S_{system} + \Delta S_{surroundings} > 0$ **TD0036**

7. If $\Delta H_{vaporisation}$ of substance X(ℓ) (molar mass = 30 g/mol) is 300 J/g at it's boiling point 300 K, then molar entropy change for reversible condensation process is :-

(A) 30 J/mol.K (B) –300 J/mol.K (C) –30 J/mol.K (D) –10 J/mol.K

TD0037

8. For 1 mole of an ideal monoatomic gas on moving from one state to other, the temperature is doubled but pressure becomes $\sqrt{2}$ times. Then entropy change in the process will be (R = 2 Cal/mol-K)

(A) R ln 2 (B) 2R ln 2 (C) 3R ln 2 (D) $\frac{R}{2} ln 2$ **TD0038**

9. If S^0 for H_2, Cl_2 and HCl are 0.13, 0.22 and 0.19 kJ K^{-1} mol^{-1} respectively. The total change in standard entropy for the reaction, $H_2 + Cl_2 \longrightarrow 2HCl$ is :

(A) 30 JK^{-1} mol^{-1} (B) 40 JK^{-1} mol^{-1} (C) 60 JK^{-1} mol^{-1} (D) 20 JK^{-1} mol^{-1}

TD0039

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10. The entropy change when two moles of ideal monoatomic gas is heated from 200°C to 300°C reversibly and isochorically :-

(A) $\frac{3}{2}R\ln\left(\frac{300}{200}\right)$ (B) $\frac{5}{2}R\ln\left(\frac{573}{273}\right)$ (C) $3R\ln\left(\frac{573}{473}\right)$ (D) $\frac{3}{2}R\ln\left(\frac{573}{473}\right)$

TD0133

11. 5 mole of an ideal gas expands reversibly from a volume of 8 dm^3 to 80dm^3 at a constant temperature of 27°C. The change in entropy is :–

(A) 41.57 JK^{-1} (B) – 95.73 JK^{-1} (C) 95.73 JK^{-1} (D) – 41.57 JK^{-1}

TD0134

12. Change in entropy is negative for :-

(A) Bromine (ℓ) $\longrightarrow$ Bromine (g) (B) $C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)$

(C) $N_2(g,10atm,298\ K) \longrightarrow N_2(g,1atm,298K)$ (D) Fe(at 400 K) $\longrightarrow$ Fe(at 300 K)

TD0135

13. For a perfectly crystalline solid $C_{p,m} = aT^3$, where a is constant. If $C_{p,m}$ is 0.42 J/K-mol at 10 K, molar entropy at 10 K is

(A) 0.42 J/K-mol (B) 0.14 J/K-mol (C) 4.2 J/K-mol (D) Zero

TD0040

14. Identify the correct statement regarding entropy.

(A) At absolute zero, the entropy of perfectly crystalline substance is taken to be +ve

(B) At absolute zero, entropy of perfectly crystalline substance is taken to be zero.

(C) At 0 °C, the entropy of a perfectly crystalline substance is taken to be zero.

(D) At absolute zero, the entropy of all crystalline substances is taken to be zero. **TD0041**

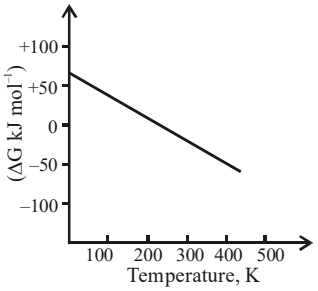
15. In which of the following reactions do you expect to have a decrease in entropy?

(A) $Fe(s) \longrightarrow Fe(l)$ (B) $2\ Fe(s) + 3/2\ O_2(g) \longrightarrow Fe_2O_3(s)$

(C) $HF(l) \longrightarrow HF(g)$ (D) $2\ H_2O_2(l) \longrightarrow 2\ H_2O(l) + O_2(g)$

TD0042

16. What can be concluded about the values of ΔH and ΔS from this graph?



(A) $\Delta H > 0,\ \Delta S > 0$

(B) $\Delta H > 0,\ \Delta S < 0$

(C) $\Delta H < 0,\ \Delta S > 0$

(D) $\Delta H < 0,\ \Delta S < 0$

TD0043

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17. For the reaction at 300 K

$$\mathbf{A(g) + B\ (g) \longrightarrow C\ (g)}$$

$\Delta U = -3.0$ kcal ; $\Delta S = -10.0$ cal/K

value of ΔG is ?

(A) –600 cal (B) –6600 cal (C) –6000 cal (D) none **TD0044**

18. For the reaction that taking place at certain temperature $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, if equilibrium pressure is X bar, then $\Delta_r G°$ would be :-

(A) – 2 RT ln X (B) – RT (ln X – ln2)

(C) – 2 RT (ln X–ln 2) (D) – 0.5 RT (ln X–ln 2) **TD0045**

19. What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure ?

(A) 80 cal (B) 540 cal (C) 620 cal (D) zero **TD0046**

20. What is the free energy change (ΔG) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 2 atm pressure ?

(A) Zero (B) 540 cal (C) 517.13 cal (D) 510 cal **TD0047**

21. If $\Delta G° > 0$ for a gaseous reaction then :

(A) $K_P > 1$

(B) $K_P < 1$

(C) The products predominate in the equilibrium mixture

(D) Mole of product must be less than mole of reactant **TD0048**

22. Heat liberated for an ideal gas undergoing reversible isothermal process is 1200 cal at 300 K. What will be Gibb's free energy change for the process ?

(A) 1200 cal (B) –1200 cal (C) 400 cal (D) –400 cal

TD0049

23. Which of the following pairs of conditions of a chemical reaction is certain to result in a spontaneous reaction ?

(A) endothermic and decreasing disorder (B) exothermic and increasing disorder

(C) endothermic and increasing disorder (D) exothermic and decreasing disorder

TD0136

24. For a reaction at 25°C, enthalpy change (ΔH) and entropy change (ΔS) are -11.7×10^3 J mol^{-1} and –105 J mol^{-1} K^{-1} respectively. The reaction is :

(A) Spontaneous (B) Non spontaneous

(C) At equilibrium (D) Can't say anything

TD0137

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25. In conversion of lime-stone to lime,

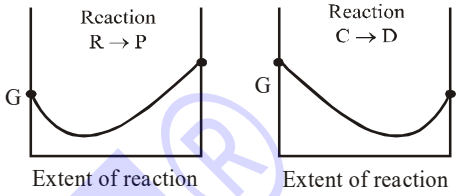
$$CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$$

the values of $\Delta H°$ and $\Delta S°$ are +179.1 kJ mol^{-1} and 160.2 J/K respectively at 298 K and 1 bar. Assuming that $\Delta H°$ and $\Delta S°$ do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is :-

(A) 1008 K (B) 1200 K (C) 845 K (D) 918 K

TD0138

26. With the help of given curves select the correct statement.



[At constant temperature & pressure]

(A) C will partially convert into D.

(B) R will not convert into P

(C) R will be completely converted into P.

(D) C will be completely converted in D.

TD0050

27. Evaluate $\Delta S°$ for the reaction below at 25°C.

$$CH_4(g) + 2Cl_2(g) \rightarrow CCl_4(\ell) + 2H_2(g)$$

Given :

	$CH_4(g)$	$Cl_2(g)$	$CCl_4(l)$	$H_2(g)$
$\Delta H°_f$ (kJ/mol)	–74.81	0	–135.4	0
$\Delta G°_f$ (kJ/mol)	–50.75	0	–65.27	0

(A) –360 J/K (B) –66.9 J/K (C) –155 J/K (D) – 487 J/K

TD0051

28. All of the following have $\Delta G°_f = 0$, except -

(A) $O_2(g)$ (B) $Br_2(g)$ (C) $H_2(g)$ (D) Ca(s)

TD0052

29. In a irreversible process taking place at constant T and P and in which only pressure-volume work is being done the change in Gibbs free energy (dG) and change in entropy (dS) satisfy the criteria:-

(A) $(dS)_{V, E} = 0, (dG)_{T, P} = 0$ (B) $(dS)_{V, E} = 0, (dG)_{T, P} > 0$

(C) $(dS)_{V, E} < 0, (dG)_{T, P} < 0$ (D) $(dS)_{V, E} > 0, (dG)_{T, P} < 0$

TD0139

30. The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_C for a solution phase reaction is -

(A) $\Delta G° = RT\ \ell nK_C$ (B) $-\Delta G° = RT\ \ell nK_C$

(C) $\Delta G = RT\ \ell nK_C$ (D) $-\Delta G = RT\ \ell nK_C$

TD0140

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EXERCISE # O-II

***Single correct* :**

1. When two equal sized pieces of the same metal at different temperatures T_h (hot piece) and T_c (cold piece) are brought into thermal contact and isolated from it's surrounding. The total change in entropy of system is given by ? Suppose heat capacity of each piece is C.

(A) $C \ln \frac{T_c + T_h}{2T_c}$ (B) $C \ln \frac{T_2}{T_1}$ (C) $C \ln \frac{(T_c + T_h)^2}{2T_h . T_c}$ (D) $C \ln \frac{(T_c + T_h)^2}{4T_h . T_c}$

TD0053

2. The change in entropy of 2 moles of an ideal gas upon isothermal expansion at 243.6 K from 20 litre until the pressure becomes 1 atm, is :-

(A) 1.385 cal/K (B) – 1.2 cal/K (C) 1.2 cal/K (D) 2.77 cal/K

TD0054

3. For the hypothetical reaction,

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

$\Delta_r G$ and $\Delta_r S$ are 20 kJ/mole and $-20\ JK^{-1}\ mol^{-1}$ respectively at 200 K. If $\Delta_r C_p$ is $20\ JK^{-1}\ mol^{-1}$ then $\Delta_r H$ at 400 K is :-

(A) 20 kJ/mole (B) 7.98 kJ/mole (C) 28 kJ/mole (D) 16 kJ/mole

TD0055

4. 1 mole of ice at 0ºC is converted in steam at 100ºC then calculated ΔS, in the process. enthalpy of vapourisation and fusion are 540 cal gm^{-1} and 80 cal gm^{-1} respectively. Use the average heat capacity of liquid water as 1cal gm^{-1} degree.

(A) $18\left(\frac{80}{373} + \frac{540}{373} + ln\frac{373}{273}\right)$ (B) $18\left(\frac{80}{273} + \frac{540}{373} + ln\frac{373}{273}\right)$

(C) $18\left(\frac{80}{273} + \frac{540}{373} + ln\frac{273}{373}\right)$ (D) $18\left(\frac{80}{273} + \frac{540}{373} + 100\right)$ **TD0056**

5. The value of ΔG_f° of gaseous mercury is 38 kJ/mole. At what total external pressure mercury start boiling at 27°C. (R.ln10 = 19 J/K-mol)

(A) $10^{-6.67}$ (B) $10^{-1.67}$ (C) $10^{-13.33}$ (D) $10^{-3.33}$ **TD0057**

6. If molar internal energy for a gas in a closed rigid vessel given by

$$U = a + bT + cT^2$$

find the entropy change (in J/K) at constant volume when 1 mol of gas are heated from 200 K to 400K.

[Given : a = 20 J/mol;b = 10 J/K-mol;c = 2×10^{-2} J/K^2-mol] (*ln* 2 = 0.7)

(A) 15 (B) 30 (C) 60 (D) 9.78 **TD0058**

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7. **Statement -1 :** When process H_2O (s) $\rightleftharpoons$ H_2O (l), reaches equiilibrium in a closed system at constant temperature and pressure, Gibb's function of H_2O (s) & H_2O (l) become same.
Statement - 2 : For reversible phase change at constant temperature & pressure, change in Gibb's free energy will be zero.
(A) Statement-1 is true, statement- 2 is true and statement- 2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false
(D) Statement-1 is false, statement-2 is true. **TD0059**

8. **Statement- 1 :** Absolute entropy of an ion in aqueous solution at 298 K may be negative.
Statement -2 : Absolute entropy at 298 K of any substance can never be negative.
(A) Statement-1 is true, statement- 2 is true and statement- 2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false
(D) Statement-1 is false, statement-2 is true. **TD0060**

9. **Statement-1 :** Net heat absorbed in a cyclic process must be always equal to net work done by the system in the cyclic process.
Statement-2 : Internal energy of system is a function of state.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **TD0061**

10. **Statement-1 :** Entropy change in reversible adiabatic expansion of an ideal gas is zero.
Statement-2 : The increase in entropy due to volume increase just compensate the decrease in entropy due to fall in temperature.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **TD0062**

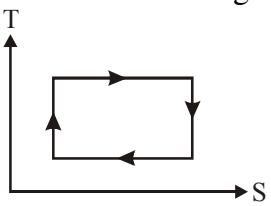
MORE THAN ONE MAY BE CORRECT :

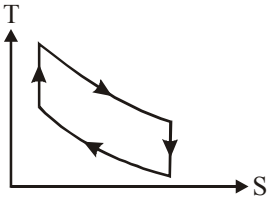
11. Select the correct statement(s).
(A) In a reversible process, ΔG is always zero in a closed system.
(B) In a reversible process, ΔS_{univ} is always zero in a closed system.
(C) In a reversible process, ΔS_{sys} is always zero in a closed system.
(D) In a reversible process, ΔS_{sys} is always zero in an isolated system. **TD0063**

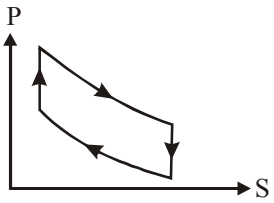
12. Which of the following statement (s) is/ are correct ?
(A) The quantities E, H and G have the same dimension
(B) Gibb's free energy of 10 gm ice at 0° C and 1.0 atm is less than the Gibb's free energy of 10 gm water at 0° C and 1 atm.
(C) $\Delta S_{sys} = 0$ for every adiabatic process in a closed system.
(D) For every reversible process in a closed system, $\Delta S_{sys} = \frac{\Delta H_{sys}}{T}$ **TD0064**

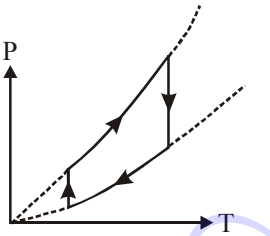
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13. Which of the following represents the carnot cycle-

(A) 

(B) 

(C) 

(D) 

TD0065

14. The normal boiling point of a liquid 'A' is 350 K. ΔH_{vap} at normal boiling point is 35 kJ/mole. Pick out the correct statement(s). (Assume ΔH_{vap} to be independent of pressure).

(A) $\Delta S_{vaporisation} > 100$ J/K mole at 350 K and 0.5 atm

(B) $\Delta S_{vaporisation} < 100$ J/K mole at 350 K and 0.5 atm

(C) $\Delta S_{vaporisation} < 100$ J/K mole at 350 K and 2 atm

(D) $\Delta S_{vaporisation} = 100$ J/K mole at 350 K and 2 atm **TD0066**

15. In isothermal ideal gas compression :

(A) w is + ve (B) ΔH is zero (C) ΔS_{gas} is + ve (D) ΔG is + ve

TD0067

16. Which of the following statement (s) is/are false :

(A) $\Delta_r S$ for $\frac{1}{2}N_2(g) \longrightarrow N(g)$ is positive

(B) ΔG_{system} is always zero for a reversible process in a closed system

(C) $\Delta G°$ for an ideal gas is a function of temperature and pressure

(D) Entropy of a closed system is always maximized at equilibrium **TD0068**

17. Which of the following processes are spontaneous ?

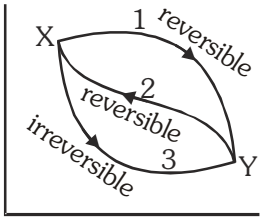
(A) Burning of fossil fuel

(B) Decomposition of water into H_2 and O_2 gas at room temperature

(C) Spreading of perfume in a room

(D) Diffusion of gas from high pressure to low pressure **TD0069**

18. Suppose a system make a transition from state X to state Y.



Given : $\Delta S_{XY} = 10$ J/K

(A) The state Y is more disordered than state X. (B) ΔS_{XY} for path 1 and 3 is same.

(C) $\Delta S_{YX} = -10$ J/K (D) The transition X → Y must be spontaneous.

TD0070

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19. In which of the following processes involving an ideal gas, entropy of system remains constant?

(A) Reversible isothermal expansion (B) Irreversible adiabatic expansion

(C) Reversible adiabatic expansion (D) Free expansion **TD0071**

20. In which of the following process involving ideal gas, entropy of surrounding remains constant ?

(A) Reversible isobaric heating (B) Reversible adiabatic expansion

(C) Irreversible adiabatic compression (D) Free expansion **TD0072**

Paragraph for Question 21 & 22

The reactions whose ΔG° are positive can not take place under standard state conditions. However another reaction whose ΔG° is negative can be coupled with the former type of reaction to give overall spontaneous process.

Consider the given reactions whose ΔG° at 300 K are provided to answer following questions.

$A(g) + 3B(g) \longrightarrow 2C(g)$: $\Delta G^\circ = +24$ kJ/mol ; $\Delta H^\circ = -24$ kJ/mol

$3C(g) \longrightarrow 2D(g)$: $\Delta G^\circ = -60$ kJ/mol ; $\Delta H^\circ = -84$ kJ/mol

All data at 300 K temperature.

21. What is ΔS° at 300 K of reaction (in J/Kmole) :

$3A(g) + 9B(g) \longrightarrow 4D(g)$

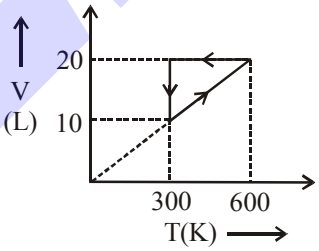
(A) –640 (B) –480 (C) +640 (D) –240 **TD0073**

22. Assuming ΔH° and ΔS° do not vary with temperature. At what minimum temperature reaction $A(g) + 3B(g) \longrightarrow 2C(g)$ become spontaneous-

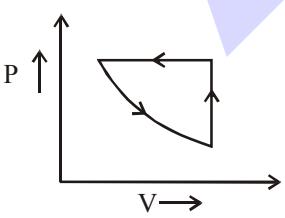
(A) 6250 K (B) 625 K (C) 150 K (D) 1000 K **TD0074**

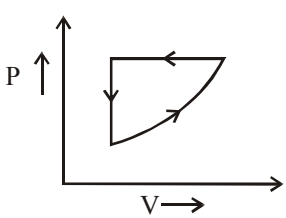
Paragraph for Q.23 & Q.24

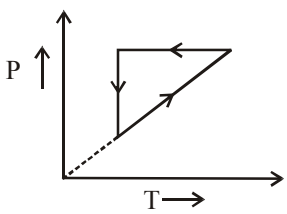
One mol of ideal monoatomic gas undergo the state change as shown in the following graph
(ln 2 = 0.7)

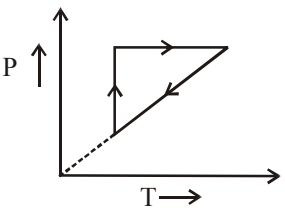


23. Correct graph for the process in paragraph is - **TD0075**

(A) 

(B) 

(C) 

(D) 

24. Efficiency of the cycle will be -

(A) 50% (B) 25 % (C) 13.63 % (D) 12 % **TD0076**

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Table type question :

Column-I	Column-II	Column-III
(A) Combustion of butane gas at S.T.P.	(P) Entropy change of universe constant	(I) Low temperature conditions favourable
(B) Dissolution of sodium chloride in water at STP	(Q) Entropy change of system positive	(II) More spontaneous at high temperature
(C) Adding carbon dioxide gas to water to form carbonated water at high pressure and 298K	(R) Enthalpy change of system negative	(III) High pressure condition favourable
(D) Vaporisation of liquid water at 1 atm and 373K	(S) Entropy change of universe positive	(IV) More spontaneous at low pressure

25. Which of following option is incorrect -

(A) A – S – I (B) B – Q – IV (C) C – R – I (D) D – P – IV

TD0077

26. In column-I (A) if the reaction was carried at 400 K which option would be correct -

(A) A – P – II (B) A – Q – III (C) A – S – IV (D) A – R – III

TD0078

27. In column-I (D), if vaporisation is carried out at 15 atm which option is correct -

(A) D – P – IV (B) D – S – II (C) D – R – I (D) D – Q – II

TD0079

MATCH THE LIST :

28. Match the List :

List-I		List-II	
(P)	H_2O (ℓ, 1 atm, 363 K) $\rightarrow$ H_2O (g, 1 atm, 363 K)	(1)	$\Delta_r S > 0$
(Q)	H_2O (s, 1 atm, 373 K) $\rightarrow$ H_2O (g, 1 atm, 373 K)	(2)	$\Delta_r G > 0$
(R)	H_2O (ℓ, 1 atm, 273 K) $\rightarrow$ H_2O (s, 1 atm, 273 K)	(3)	$\Delta_r H < 0$
(S)	H_2O (s, 1 atm, 353 K) $\rightarrow$ H_2O (ℓ, 1 atm, 353 K)	(4)	$\Delta_r U > 0$

TD0080

Code:

	P	Q	R	S
(A)	2	4	1	3
(B)	4	2	3	1
(C)	2	1	3	4
(D)	4	3	1	2

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MATCH THE COLUMN :

29.

Column–I	**Column-II**
(A) Reversible adiabatic compression	(P) $\Delta S_{system} > 0$
(B) Reversible vaporisation	(Q) $\Delta S_{system} < 0$
(C) Adiabatic free expansion of ideal gas in vacuum	(R) $\Delta S_{surroundig} < 0$
(D) Dissociation of $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$	(S) $\Delta S_{surrounding} = 0$

TD0081

30.

Column-I **(Related to process)**	**Column-II** **(Related to system)**
(A) Fusion at melting point	(P) $\Delta G = 0$
(B) Vapourisation at boiling point	(Q) $\Delta G < 0$
(C) Condensation at triple point	(R) $\Delta S > 0$
(D) Melting at normal boiling point	(S) $\Delta H \simeq \Delta U$

TD0082

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EXERCISE # J-MAIN

1. The entropy (S°) of the following substances are : **[JEE-MAINS-(online) 2014]**

CH_4 (g) 186.2 J K^{-1} mol^{-1}

O_2 (g) 205.0 J K^{-1} mol^{-1}

CO_2 (g) 213.6 J K^{-1} mol^{-1}

H_2O (l) 69.9 J K^{-1} mol^{-1}

The entropy change ($\Delta S°$) for the reaction

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O$ (ℓ) is:-

(1) –312.5 JK^{-1} mol^{-1} (2) – 37.6 JK^{-1} mol^{-1} (3) – 108.1 JK^{-1} mol^{-1} (4) – 242.8 JK^{-1} mol^{-1}

TD0085

2. The molar heat capacity (C_p) of CD_2O is 10 cals at 1000 K. The change in entropy associated with cooling of 32 g of CD_2O vapour from 1000 K to 100 K at constant pressure will be

(D = deuterium, at. mass = 2u) **[JEE-MAINS-(online) 2014]**

(1) – 23.03 cal deg^{-1} (2) 2.303 cal deg^{-1} (3) 23.03 cal deg^{-1} (4) – 2.303 cal deg^{-1}

TD0086

3 $\Delta_f G°$ at 500 K for substance 'S' in liquid state and gaseous state are + 100.7 kcal mol^{-1} and + 103 kcal mol^{-1}, respectively. Vapour pressure of liquid 'S' at 500 K is approximately equal to :

(R = 2 cal K^{-1} mol^{-1}) - **[JEE-MAINS-(online) 2018]**

(1) 0.1 atm (2) 10 atm (3) 100 atm (4) 1 atm **TD0087**

4. For which of the following processes, ΔS is negative? **[JEE-MAINS-(online) 2018]**

(1) C(diamond) $\rightarrow$ C(graphite) (2) N_2(g, 273 K) $\rightarrow$ N_2(g, 300K)

(3) $H_2(g) \rightarrow 2H(g)$ (4) N_2(g, 1 atm) $\rightarrow$ N_2(g, 5 atm) **TD0088**

5. At 320 K, a gas A_2 is 20% dissociated to A(g). The standard free energy change at 320 K and 1 atm in J mol^{-1} is approximately : **[JEE-MAINS-(online) 2018]**

(R = 8.314 JK^{-1} mol^{-1}; ℓn 2 = 0.693; ℓn 3 = 1.098)

(1) 4281 (2) 4763 (3) 2068 (4) 1844 **TD0089**

6. Two blocks of the same metal having same mass and at temperature T_1 and T_2, respectively. are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS, for this process is : **[JEE-MAINS-(online) 2019]**

(1) $2C_P \ln\left(\frac{T_1+T_2}{4T_1T_2}\right)$ (2) $2C_P \ln\left[\frac{(T_1+T_2)^{\frac{1}{2}}}{T_1T_2}\right]$ (3) $C_P \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$ (4) $2C_P \ln\left[\frac{T_1+T_2}{2T_1T_2}\right]$

TD0090

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7. For the chemical reaction $X \rightleftharpoons Y$, the standard reaction Gibbs energy depends on temperature T (in K) as : **[JEE-MAINS-(online) 2019]**

$$\Delta_r G^o \text{ (in kJ mol}^{-1}) = 120 - \frac{3}{8}T$$

The major component of the reaction mixture at T is :

(1) X if T = 315 K (2) X if T = 350 K (3) Y if T = 300 K (4) Y if T = 280 K

TD0091

8. For the equilibrium, **[JEE-MAINS-(online) 2019]**

$2H_2O \rightleftharpoons H_3O^+ + OH^-$, the value of ΔG^o at 298 K is approximately :-

(1) –80 kJ mol^{-1} (2) –100 kJ mol^{-1} (3) 100 kJ mol^{-1} (4) 80 kJ mol^{-1}

TD0092

9. The standard reaction Gibbs energy for a chemical reaction at an absolute temperature T is given by

$$\Delta_r G^o = A - BT$$

Where A and B are non-zero constants. Which of the following is TRUE about this reaction ?

(1) Exothermic if B < 0 **[JEE-MAINS-(online) 2019]**

(2) Exothermic if A > 0 and B < 0

(3) Endothermic if A < 0 and B > 0

(4) Endothermic if A > 0 **TD0093**

10. The reaction, MgO(s) + C(s)→Mg(S) + CO(g), for which $\Delta_r H^o$ = + 491.1 kJ mol^{-1} and $\Delta_r S^o$ = 198.0 JK^{-1} mol^{-1}, is not feasible at 298 K. Temperature above which reaction will be feasible is :- **[JEE-MAINS-(online) 2019]**

(1) 1890.0 K (2) 2480.3 K (3) 2040.5 K (4) 2380.5 K

TD0094

11. A process has ΔH = 200 Jmol^{-1} and

ΔS = 40 JK^{-1}mol^{-1}. Out of the values given below, choose the minimum temperature above which the process will be spontaneous : **[JEE-MAINS-(online) 2019]**

(1) 5 K (2) 4 K (3) 20 K (4) 12 K **TD0095**

12. The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapours at 383 K is : **[JEE-MAINS-(online) 2019]**

(Specific heat of water liquid and water vapour are 4.2 kJ K^{-1} kg^{-1} and 2.0 kJ K^{-1} kg^{-1} ; heat of liquid fusion and vapourisation of water are 344 kJ kg^{-1} and 2491 kJ kg^{-1}, respectively).

(log 273 = 2.436, log 373 = 2.572, log 383 = 2.583)

(1) 7.90 kJ kg^{-1} K^{-1} (2) 2.64 kJ kg^{-1} K^{-1} (3) 8.49 kJ kg^{-1} K^{-1} (4) 9.26 kJ kg^{-1} K^{-1}

TD0096

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13. The process with negative entropy change is : **[JEE-MAINS-(online) 2019]**

(1) Dissolution of iodine in water

(2) Synthesis of ammonia from N_2 and H_2

(3) Dissolution of $CaSO_4(s)$ to $CaO(s)$ and $SO_3(g)$

(4) Subimation of dry ice **TD0097**

14. For the reaction ; **[JEE-MAINS-(online) 2020]**

$A(l) \rightarrow 2B(g)$

$\Delta U = 2.1$ kcal , $\Delta S = 20$ cal K^{-1} at 300 K

Hence ΔG in kcal is_____ . **TD0098**

15. The true statement amongst the following is: **[JEE-MAINS-(online) 2020]**

(1) Both ΔS and S are functions of temperature.

(2) S is not a function of temperature but ΔS is a function of temperature.

(3) Both S and ΔS are not functions of temperature.

(4) S is a function of temperature but ΔS is not a function of temperature.

TD0141

16. For the reaction $A_{(g)} \rightarrow (B)_{(g)}$, the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of $\Delta_r G$ for the reaction at 300 K and 1 atm in J mol^{-1} is – xR, where x is _____ (Rounded of to the nearest integer) ($R = 8.31$ J mol^{-1} K^{-1} and $\ln 10 = 2.3$) **[JEE-MAINS-(online) 2020]**

TD0142

17. Assuming ideal behaviour, the magnitude of log K for the following reaction at 25°C is $x \times 10^{-1}$. The value of x is _____. (Integer answer) **[JEE-MAINS-(online) 2021]**

$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_{6(\ell)}$

[Given: $\Delta_f G^o(HC\equiv CH) = -2.04 \times 10^5$ J mol^{-1} ; $\Delta_f G^o(C_6H_6) = -1.24 \times 10^5$ J mol^{-1}; $R = 8.314$ J K^{-1} mol^{-1}]

TD0143

18. For a chemical reaction $A + B \rightleftharpoons C + D$ **[JEE-MAINS-(online) 2021]**

($\Delta_r H^0 = 80$kJ mol^{-1}) the entropy change $\Delta_r S^0$ depends on the temperature T (in K) as ($\Delta_r S^0 = 2T$ (J K^{-1} mol^{-1}).

Minimum temperature at which it will become spontaneous is _____K.(Integer)

TD0144

19. For the reaction $A(g) \rightleftharpoons B(g)$ at 495 K, $\Delta_r G^o = -9.478$ kJ mol^{-1}. **[JEE-MAINS-(online) 2021]**

If we start the reaction in a closed container at 495 K with 22 millimoles of A, the amount of B in the equilibrium mixture is _____ millimoles. (Round off to the Nearest Integer).

[$R = 8.314$ J mol^{-1} K^{-1}; $\ell n\ 10 = 2.303$]

TD0145

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20. For a given chemical reaction A $\rightarrow$ B at 300 K the free energy change is -49.4 kJ mol^{-1} and the enthalpy of reaction is 51.4 kJ mol^{-1}. The entropy change of the reaction is _____ J K^{-1} mol^{-1}.

[JEE-MAINS-(online) 2021]

TD0146

21. Data given for the following reaction is as follows: **[JEE-MAINS-(online) 2021]**

$FeO(s) + C(graphite) \rightarrow Fe(s) + CO(g)$

Substance	ΔH° ($kJ\ mol^{-1}$)	ΔS° ($J\ mol^{-1}K^{-1}$)
$FeO_{(s)}$	–266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe_{(s)}$	0	27.28
$CO_{(g)}$	–110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is _____ .
(Integer answer)

TD0147

22. For the reaction $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when **[JEE-MAINS-(online) 2021]**
$\Delta S = -176.0\ JK^{-1}$ and $\Delta H = -57.8\ kJ\ mol^{-1}$, the magnitude of ΔG at 298 K for the reaction is_____ kJ mol^{-1}. (Nearest integer).

TD0148

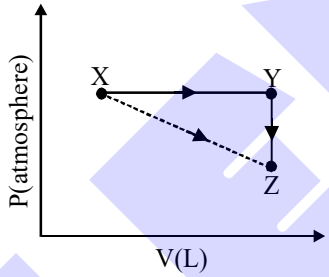
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EXERCISE # J-ADVANCED

1. Match the transformations in **Column-I** with appropriate option in **Column-II** **[JEE 2011]**

	Column-I		**Column-II**
(A)	$CO_2(s) \rightarrow CO_2(g)$	(p)	phase transition
(B)	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$	(q)	allotropic change
(C)	$2H^{\bullet} \rightarrow H_2(g)$	(r)	ΔH is positive
(D)	$P_{(white,\ solid)} \rightarrow P_{(red,solid)}$	(s)	ΔS is positive
		(t)	ΔS is negative **TD0099**

2. For an ideal gas, consider only P-V work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct ? [take ΔS as change in entropy and w as work done]**[JEE 2012]**



(A) $\Delta S_{x\rightarrow z} = \Delta S_{x\rightarrow y} + \Delta S_{y\rightarrow z}$ (B) $W_{x\rightarrow z} = W_{x\rightarrow y} + W_{y\rightarrow z}$

(C) $W_{x\rightarrow y\rightarrow z} = W_{x\rightarrow y}$ (D) $\Delta S_{x\rightarrow y\rightarrow z} = \Delta S_{x\rightarrow y}$ **TD0100**

3. For the process **[JEE 2014]**

$$H_2O(l) \rightarrow H_2O(g)$$

at T = 100°C and 1 atmosphere pressure, the correct choice is

(A) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} > 0$ (B) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} < 0$

(C) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} > 0$ (D) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} < 0$ **TD0101**

4. Match the thermodynamic processes given under Column-I with the expressions given under Column-II. **[JEE 2015]**

	Column-I		**Column - II**
(A)	Freezing of water at 273 K and 1 atm	(P)	q = 0
(B)	Expansion of 1 mol of an ideal gas into a vacuum under isolated conditions	(Q)	w = 0
(C)	Mixing of equal volumes of two ideal gases at constant temeprature and pressure in an isolated container	(R)	$\Delta S_{sys} < 0$
(D)	Reversible heating of $H_2(g)$ at 1 atm from 300 K to 600 K followed by reversible cooling to 300 K at 1 atm	(S)	$\Delta U = 0$
		(T)	$\Delta G = 0$ **TD0102**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1 Eq-(RC)\Eng\3-Thermodynamic-02\02_Ex..p65

5. One mole of an ideal gas at 300 K in thermal contact with surroundings expands isothermally from 1.0 L to 2.0 L against a constant pressure of 3.0 atm. In this process, the change in entropy of surroundings (ΔS_{surr}) in J K^{-1} is - **[JEE 2016]**

(1 L atm = 101.3 J)

(A) 5.763 (B) 1.013 (C) –1.013 (D) –5.763 **TD0103**

Paragraph for Q.6 & Q.7

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation : **[JEE 2016]**

$$X_2(g) \rightleftharpoons 2X(g)$$

The standard reaction Gibbs energy, $\Delta_r G°$, of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β . Thus, $\beta_{equilibrium}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally.

(Given : R = 0.083 L bar K^{-1} mol^{-1})

6. The equilibrium constant K_P for this reaction at 298 K, in terms of $\beta_{equilibrium}$, is

(A) $\frac{8\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ (B) $\frac{8\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$ (C) $\frac{4\beta^2_{equilibrium}}{2-\beta_{equilibrium}}$ (D) $\frac{4\beta^2_{equilibrium}}{4-\beta^2_{equilibrium}}$

TD0104

7. The **INCORRECT** statement among the following for this reaction is

(A) Decrease in the total pressure will result in formation of more moles of gaseous X

(B) At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously

(C) $\beta_{equilibrium} = 0.7$

(D) $K_C < 1$ **TD0105**

8. The standard state Gibbs free energies of formation of C(graphite) and C(diamond) at T = 298 K are **[JEE 2017]**

$$\Delta_f G° \text{ [C(graphite)]} = 0 \text{ kJ mol}^{-1}$$

$$\Delta_f G° \text{ [C(diamond)]} = 2.9 \text{ kJ mol}^{-1}$$

The standard state means that the pressure should be 1 bar , and substance should be pure at a given temperature. The conversion of graphite [C(graphite)] to diamond [C(diamond)] reduces its volume by 2×10^{-6} m^3 mol^{-1}. If C(graphite) is converted to C(diamond) isothermally at T = 298 K, the pressure at which C(graphite) is in equilibrium with C(diamond), is

[Useful information : 1 J = 1 kg m^2 s^{-2} ; 1 Pa = 1 kg m^{-1} s^{-2} ; 1 bar = 10^5 Pa]

(A) 14501 bar (B) 29001 bar (C) 58001 bar (D) 1405 bar **TD0106**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1-Eq-(PC)\Eng\3-Thermodynamic-02\02_Ex_p65

9. For a reaction taking place in a container in equilibrium with its surroundings, the effect of temperature on its equilibrium constant K in terms of change in entropy is described by

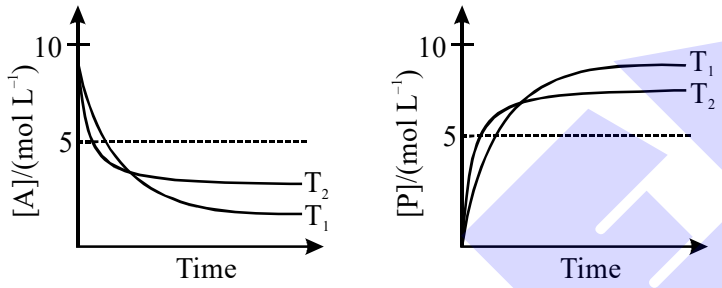
(A) With increase in temperature, the value of K for exothermic reaction decreases because the entropy change of the system is positive

(B) With increase in temperature, the value of K for endothermic reaction increases because unfavourable change in entropy of the surroundings decreases

(C) With increase in temperature, the value of K for exothermic reaction decreases because favourable change in entropy of the surroundings decreases

(D) With increase in temperature, the value of K for endothermic reaction increases because the entropy change of the system negative **TD0107**

10. For a reaction, A $\rightleftharpoons$ P, the plots of [A] and [P] with time at temperatures T_1 and T_2 are given below. **[JEE 2018]**



If $T_2 > T_1$, the correct statement(s) is (are)

(Assume ΔH^θ and ΔS^θ are independent of temperature and ratio of lnK at T_1 to lnK at T_2 is greater than T_2/T_1. Here H, S, G and K are enthalpy, entropy, Gibbs energy and equilibrium constant, respectively.)

(A) $\Delta H^\theta < 0$, $\Delta S^\theta < 0$ (B) $\Delta G^\theta < 0$, $\Delta H^\theta > 0$

(C) $\Delta G^\theta < 0$, $\Delta S^\theta < 0$ (D) $\Delta G^\theta < 0$, $\Delta S^\theta > 0$ **TD0108**

11. The surface of copper gets tarnished by the formation of copper oxide. N_2 gas was passed to prevent the oxide formation during heating of copper at 1250 K. However, the N_2 gas contains 1 mole % of water vapour as impurity. The water vapour oxidises copper as per the reaction given below

$2Cu(s) + H_2O(g) \rightarrow Cu_2O(s) + H_2(g)$

p_{H_2} is the minimum partial pressure of H_2 (in bar) needed to prevent the oxidation at 1250 K. The value of $\ln(p_{H_2})$ is ___.

(Given : total pressure = 1 bar, R (universal gas constant) = 8 $JK^{-1}mol^{-1}$, ln(10) = 2.3. Cu(s) and $Cu_2O(s)$ are mutually immiscible.

At 1250 K : $2Cu(s) + 1/2O_2(g) \rightarrow Cu_2O(s)$; $\Delta G^\theta = -78{,}000$ J mol^{-1}

$H_2(g) + 1/2O_2(g) \rightarrow H_2O(g)$; $\Delta G^\theta = -1{,}78{,}000$ J mol^{-1}; G is the Gibbs energy) **[JEE 2018]**

TD0109

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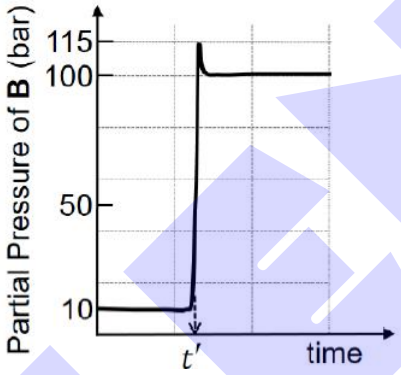
12. Consider the following reversible reaction, **[JEE 2018]**

$$A(g) + B(g) \rightleftharpoons AB(g).$$

The activition energy of the backward reaction exceeds that of the forward reaction by 2RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of ΔG^{θ} (in J mol^{-1}) for the reaction at 300 K is____.

(Given ; ln (2) = 0.7, RT = 2500 J mol^{-1} at 300 K and G is the Gibbs energy) **TD0110**

13. Consider the reaction A $\rightleftharpoons$ B at 1000 K. At time t', the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of A was maintained at 1 bar. Given below is the plot of the partial pressure of B with time. What is the ratio of the standard Gibbs energy of the reaction at 1000 K to that at 2000 K? **[JEE 2020]**



TD0149

14. Tin is obtained from cassiterite by reduction with coke. Use the data given below to determine the minimum temperature (in K) at which the reduction of cassiterite by coke would take place. **[JEE 2020]**

At 298 K : $\Delta_f H°(SnO_2(s))$ = -581.0 kJ mol^{-1}, $\Delta_f H°(CO_2(g))$ = -394.0 kJ mol^{-1}

$S°(SnO_2(s))$ = 56.0 J K^{-1} mol^{-1}, S°(Sn(s)) = 52.0 J K^{-1} mol^{-1} ,

$S°(C\ (s))$ = 6.0 J K^{-1} mol^{-1} , $S°(CO_2(g))$ = 210.0 J K^{-1} mol^{-1}.

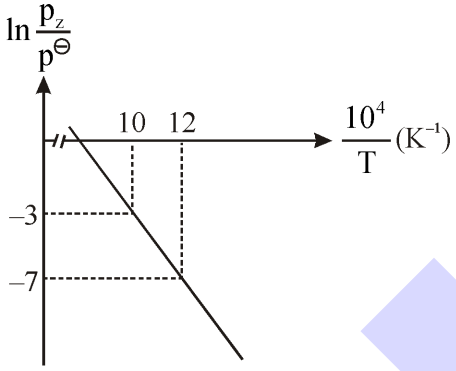
Assume that the enthalpies and the entropies are temperature independent.

TD0150

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Paragraph for Q.15 to Q.16

For the reaction X(s) $\rightleftharpoons$ Y(s) + Z(g) , the plot of ln $\frac{p_z}{p^\ominus}$ versus $\frac{10^4}{T}$ is given below (in solid line), where p_z is the pressure (in bar) of the gas Z at temperature T and $P^\ominus$ =1 bar.



(Given, $\frac{d(\ln K)}{d\left(\frac{1}{T}\right)} = -\frac{\Delta H^\ominus}{R}$ where the equilibrium constant, K = $\frac{p_z}{p^\ominus}$ and the gas constant, R = 8.314 J K^{-1} mol^{-1}) **[JEE 2021]**

15. The value of standard enthalpy, $\Delta H^\ominus$ (in kJ mol^{-1}) for the reaction is_____ .

TD0151

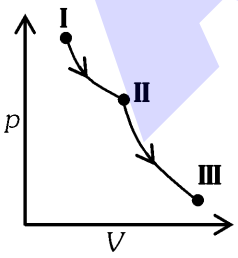
16. The value of $\Delta S^\ominus$ (in J K^{-1} mol^{-1}) for the given reaction, at 1000 K is_____ .

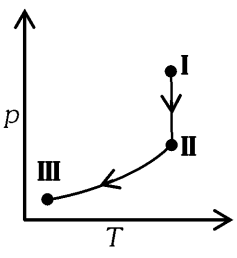
[JEE 2021]

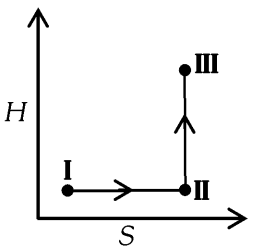
TD0152

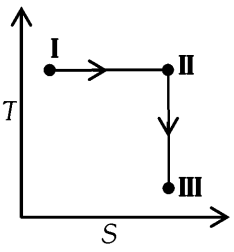
17. An ideal gas undergoes a reversible isothermal expansion from state I to state II followed by a reversible adiabatic expansion from state II to state III. The correct plot(s) representing the changes from state I to state III is(are) **[JEE 2021]**

(p : pressure, V : volume, T : temperature, H : enthalpy, S : entropy)

(A) 

(B) 

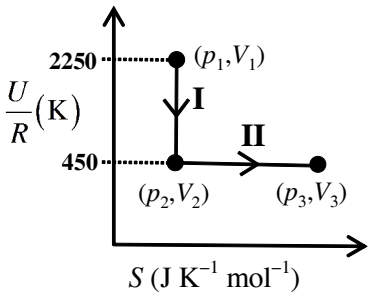
(C) 

(D) 

TD0153

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18. One mole of an ideal gas at 900 K, undergoes two reversible processes, I followed by II, as shown below. If the work done by the gas in the two processes are same, the value of $\ln\frac{V_3}{V_2}$ is ___. **[JEE 2021]**



(U: internal energy, S: entropy, p: pressure, V: volume, R: gas constant)

(Given: molar heat capacity at constant volume, $C_{v,m}$ of the gas is $\frac{5}{2}R$)

TD0154

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ANSWER KEY

EXERCISE # S-I

1. 117°C, 52°C
2. 197°C
3. (i) irrversible (ii) reversible (iii) impossible
4. 0.80
5. (4.00)
6. (i) ΔS_{syst} = 28.72 J/K ; ΔS_{surr} = –28.72 J/K ; ΔS_{total} = 0
   (ii) ΔS_{sys} = 28.72 J/K ; ΔS_{surr} = –11.22 J/K , ΔS_{total} = 17.50 J/K
7. (a) –250 J/K (b) 333.33 J/K (c) +83.33 J/K. Process is spontaneous.
8. (a) 2.10 , (b) 1.50
9. 38.20 - 38.30
10. 819.00
11. (5.00)
12. (200.00)
13. 10.32 - 10.33
14. (i) –90.5 (ii) –374.5 (iii) –3.26 (all in J mol^{-1} K^{-1})
15. [A] - II ; [B] - II ; [C] - II ; [D] - II
16. –3.70 kJ
17. 436.00 kJ
18. 5.96 – 5.97
19. (–7.00)
20. (50.00)
21. (i) $\Delta G = 0$
    (ii) $\Delta S = -5$ J/mole.K
    (iii) ΔS_{total} = + 1.66 J/ mole. K ; process in spontaneous
    (iv) ΔS_{total} = – 1 J/mole. K ; ΔG = + 250 J/mole
22. 2873.10
23. 750.00
24. –997.68 J/mol
25. ΔH° = 2.17 - 2.18 ; ΔS° = –2.08
26. 27.00
27. 0.00 , 1.00
28. –13.49
29. 2.10
30. –6

EXERCISE # S-II

1. (i) 30 kJ (ii) 60 kJ (iii) 100 J/K
2. 30 kJ/mole
3. –1745.94 J/mol; –6983.76 J/mol and 13.97 kJ/mol
4. (16.60 J/K)
5. (– 327.75 kJ/ mol)
6. (17.8)
7. (3360)
8. 10 J/K
9. 191.00 to 191.50 $JK^{-1}mol^{-1}$
10. (1195 kJ)

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse \Chem\Module\s4-TD-01-T-C-TD-02 & 1 Eq-(RC)\Eng\3-Thermodynamic-02\02_Ex..pt5

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	C	D	C	D	C	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	D	B	B	B	A	A	C	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	B	B	D	A	C	B	D	B

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	A	B	A	A	A	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	A	A,C,D	A,C	A,B,D	B,C,D	A,C,D	A,B,C	C	B,C,D
Que.	21	22	23	24	25	26	27	28		
Ans.	A	C	D	D	B	C	D	C		
Que.	29				30					
Ans.	A-S	B-P,R	C-P,S	D-P,R	A-P,S,R	B-P,R	C-P	D-Q,R,S		

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	4	2	3	1	4	4	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	2	-2.70 to -2.71	1	1380	855	200	20	336
Que.	21	22								
Ans.	964	5								

EXERCISE # J-ADVANCED

Que.	1				2	3	4			
Ans.	A-p,r,s	B-r,s	C-t	D-p,q,t	A,C	B	A-R,T	B-P,Q,S	C-P,Q,S	D-P,Q,S,T
Que.	5	6	7	8	9	10	11	12	13	14
Ans.	C	B	C	A	B,C	A,C	-14.6	8500	0.25	935
Que.	15	16	17	18						
Ans.	166.28	141.33 or 141.34	A,B,D	10						

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